

Individual differences in dynamic belief updating during trust learning



PROGRAM IN
NEUROSCIENCE &
COGNITIVE SCIENCE



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CONTEXT

- **Trust** is fundamental to social interaction. It is defined as the **willingness to be vulnerable to another being on the basis of positive expectations** of their intentions and behaviors (1).
- In repeated interactions, **learning to trust others** involves **cognitive processes** that integrate **uncertainty, context, and potential betrayal**.
- **Different strategies** can guide trust behavior. Some rely on **heuristics** (e.g., fixed rules or triggers). Others use **reinforcement learning**, adjusting expected value through associative updates, or **Bayesian belief updating**, which integrates uncertainty into probabilistic inferences about partners' intentions (2, 3).
- **Individuals differ** in how they deploy cognitive **processes**, which may impact their **strategies**. Yet, this heterogeneity has not been systematically characterized. Here, we aim to do so:
 - How does dynamic trust learning integrate sensitivity to betrayal, uncertainty and context?
 - How do these processes vary across individuals and reveal distinct cognitive profiles?

HYPOTHESES

• Gain/loss learning asymmetry

Higher learning rate after losses than gains.

• Individual differences

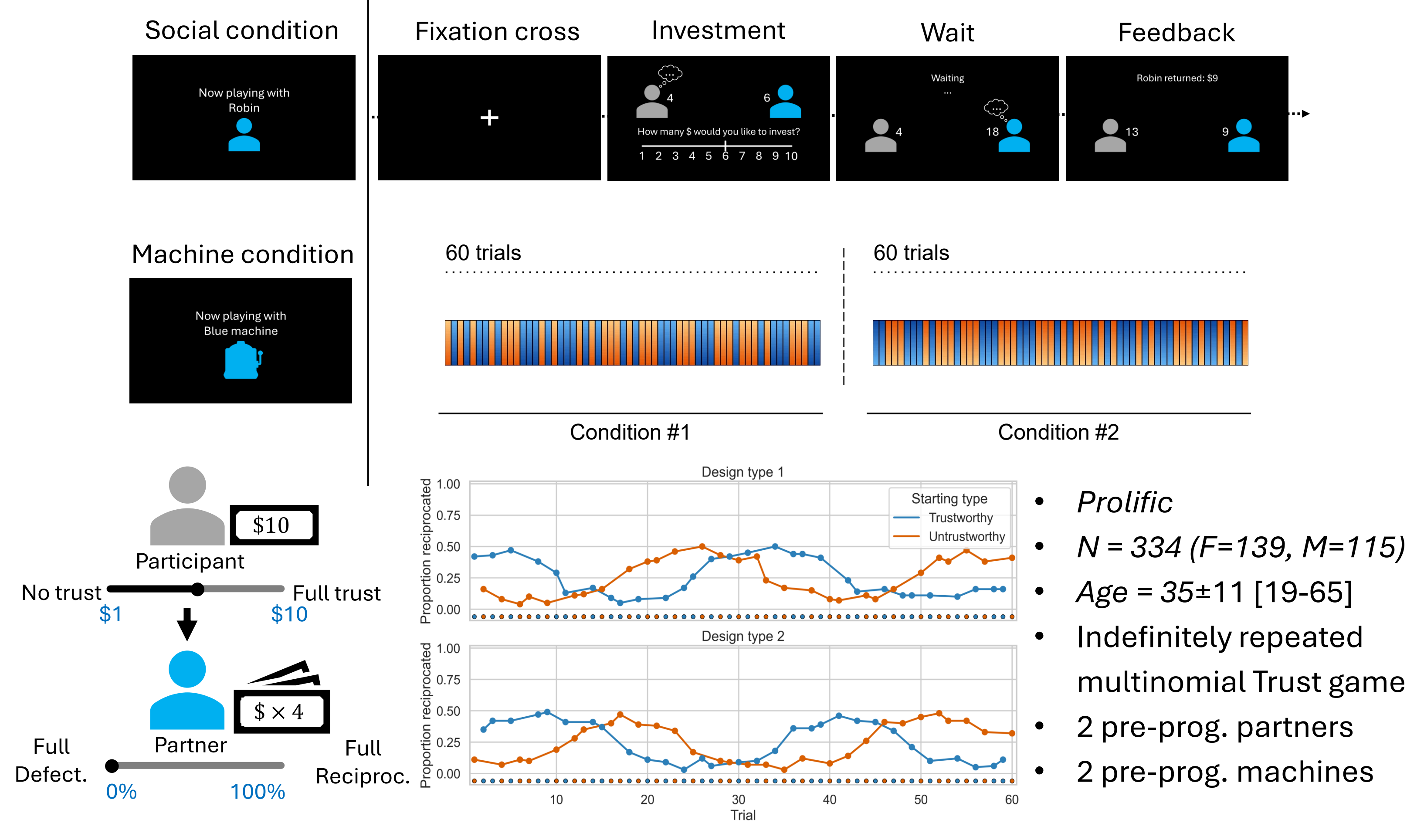
- A subset of participants deploy **heuristics** (non-learners).

• Context modulation

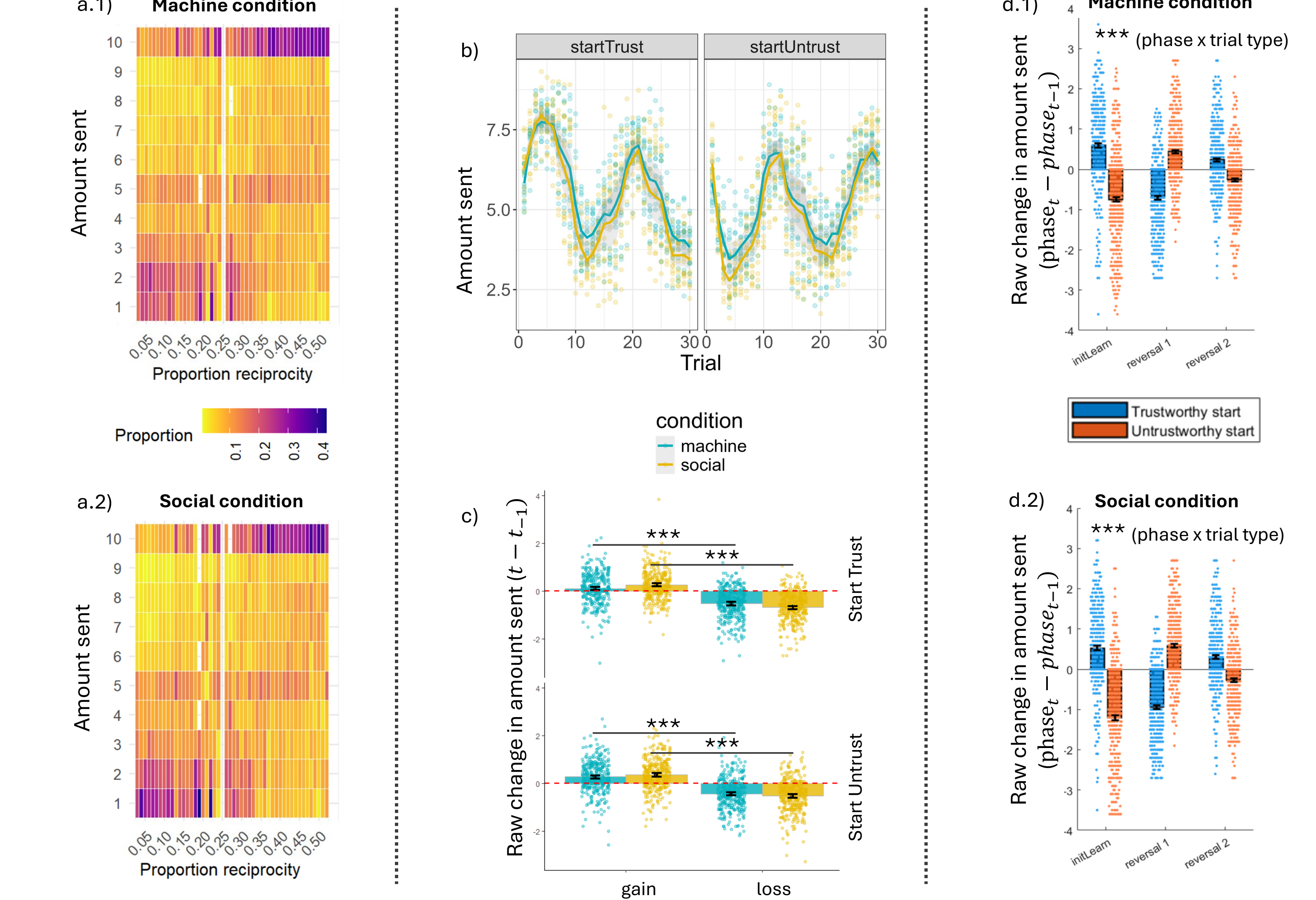
Betrayal aversion leads to **stronger learning rates after losses in the social condition**.

- Among learners, **variability** in learning asymmetry and betrayal aversion is captured by model fit/parameters

METHODS



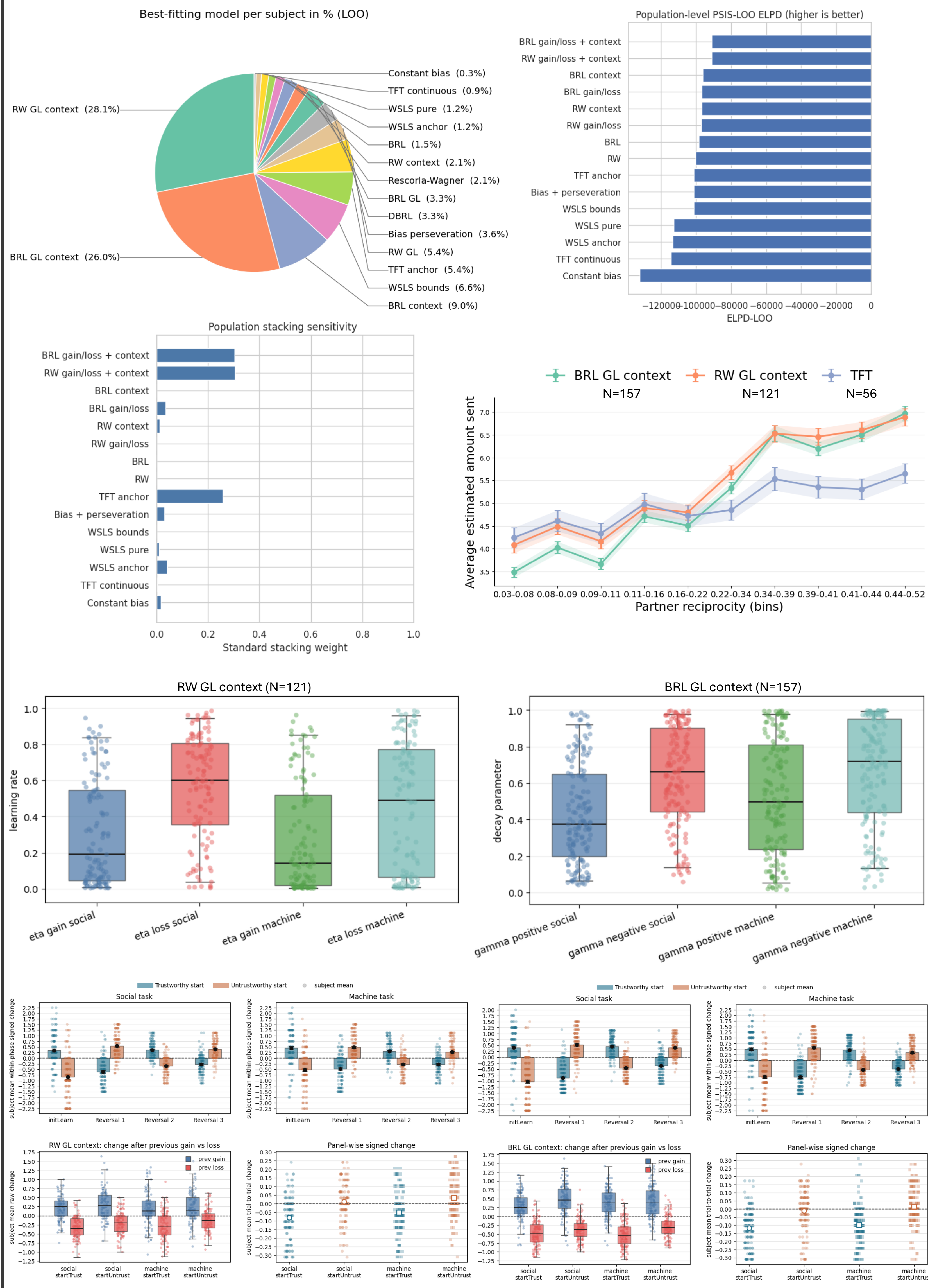
BEHAVIOR



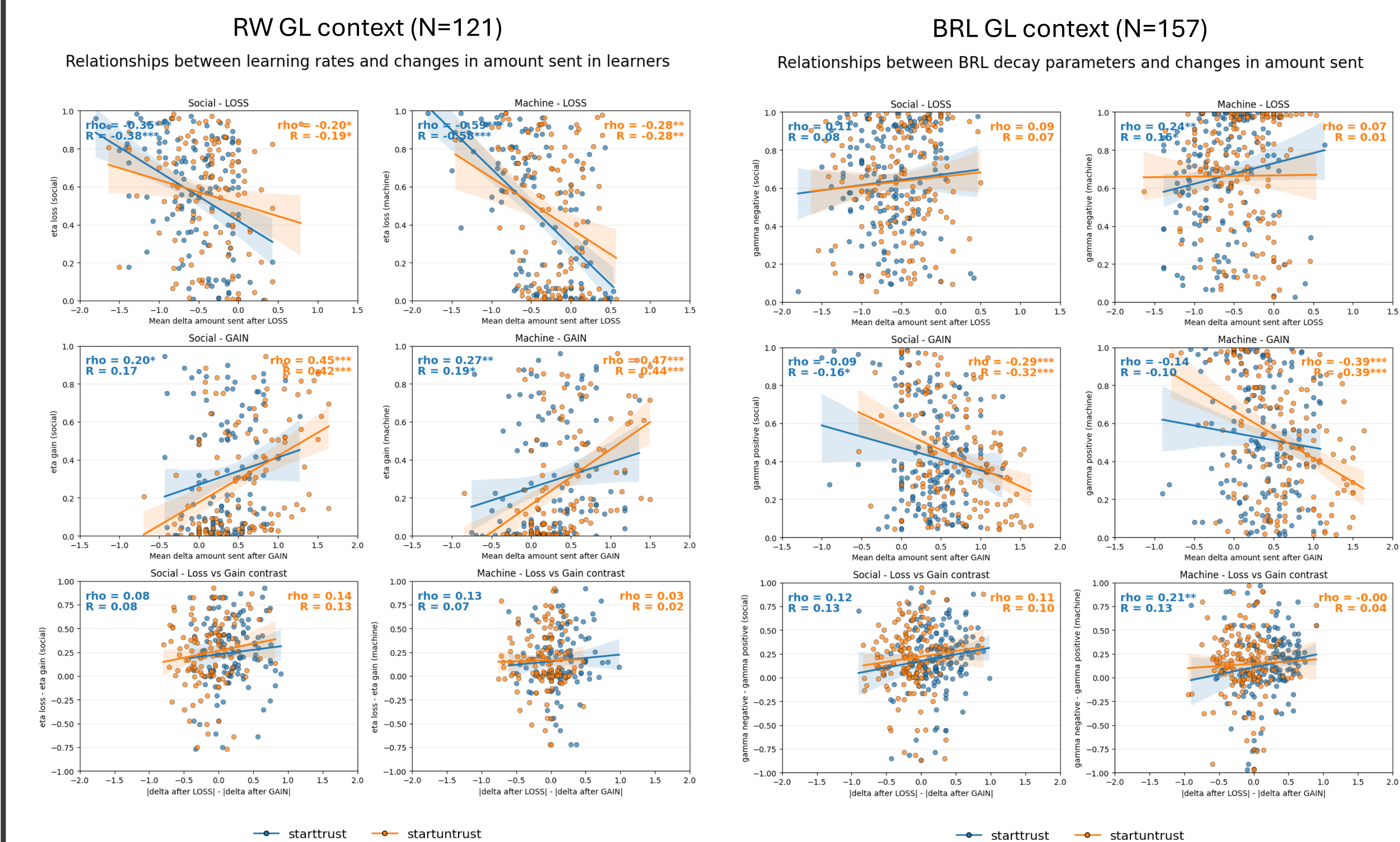
CONCLUSIONS & NEXT STEPS

- Trust behavior shows **gain/loss asymmetry, context modulation, and individual variability** in strategy use.
- **Best-fitting model was BRL GL context**, but sample is better explained by a roster of 3 models, including a heuristic.
- Among learners, the models captures different variabilities in betrayal aversion & context sensitivity. **Loss** (and gain) learning parameters correlated differently with changes in amount sent after losses (and gains) in both conditions.
- Next, we are extending the model space to include **Bayesian updating with sensitivity to uncertainty** (4).
- We will examine links between model parameters and individual-difference measures, improve the allocation method, and validate findings in a replication dataset.

GAIN-LOSS AND CONTEXT DEPENDENCY



INDIVIDUAL DIFFERENCES



REFERENCES

- (1) Alós-Ferrer, C., & Farolfi, F. (2019). *Frontiers in neuroscience*, 13, 887.
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